

75 System Design Interview Questions

A curated list of 75 system design questions with descriptions and direct links to solutions, walkthroughs, and reference architectures — primarily sourced from the **System Design Primer**, checkcheckzz/system-design-interview, and real-world engineering blogs.

Legend

- = Has a full written solution (System Design Primer)
- = Has a reference architecture or deep-dive article
- = Has a video walkthrough

The Questions

#	Question	Description	Solution / Reference
1	● Design Pastebin.com (or Bit.ly)	Design a text/code snippet sharing service with short URL generation, expiration, and high read throughput. Covers hashing, storage, CDN, and caching.	System Design Primer – Solution
2	● Design Twitter Timeline & Search	Design a social media platform that supports tweet publishing, follower feeds, and full-text search at massive scale. Covers fan-out, timelines, and search indexing.	System Design Primer – Solution
3	● Design a Web Crawler	Build a distributed system that crawls the web, respects robots.txt, deduplicates pages, and stores content for indexing. Covers politeness policies, queues, and storage.	System Design Primer – Solution
4	● Design Mint.com	Design a personal finance aggregator that pulls data from multiple bank accounts, categorizes transactions, and generates spending insights. Covers data ingestion, pipelines, and security.	System Design Primer – Solution
5	● Design Data Structures for a Social Network	Design the backend data model and graph traversal system for a social network (friend suggestions, mutual friends). Covers social graphs, BFS/DFS at scale, and sharding.	System Design Primer – Solution

6	● Design a Key-Value Store for a Search Engine	Design a low-latency query cache (key-value store) to back a search engine. Covers cache invalidation, eviction policies, and distributed consistency.	System Design Primer – Solution
7	● Design Amazon's Sales Ranking by Category	Design a feature that tracks and displays the best-selling products per category in near real-time. Covers time-windowed aggregation, batch vs. stream processing.	System Design Primer – Solution
8	● Design a System That Scales to Millions of Users on AWS	Walk through scaling a simple single-server app on AWS from zero to millions of users. Covers load balancers, auto-scaling, RDS read replicas, ElastiCache, S3, and Route 53.	System Design Primer – Solution
9	● Design a File Sync Service like Dropbox	Design a system that syncs files across devices with conflict resolution, delta sync, and offline support. Covers chunking, metadata service, and block storage.	Dropbox Engineering – How We've Scaled Dropbox (YouTube)
10	● Design a Search Engine like Google	Design a web-scale search engine with crawling, indexing, ranking, and query serving. Covers inverted index, PageRank, and distributed storage.	ACM Queue – Web Search
11	● Design a Scalable Web Crawler	Design a crawler that handles politeness, deduplication, and massive parallelism without overloading target sites. Covers URL frontier, DNS caching, and content stores.	Quora – How to build a web crawler from scratch
12	● Design Google Docs (Collaborative Editing)	Design a real-time collaborative document editing system that handles concurrent edits, conflict resolution, and history. Covers Operational Transform (OT) or CRDTs.	Neil Fraser – Differential Synchronization
13	● Design a Key-Value Store like Redis	Design an in-memory key-value store with persistence, expiry, pub/sub, and sorted sets. Covers data structures, AOF/RDB persistence, and replication.	Implementing a Key-Value Store (Blog Series)
14	● Design a Cache System like Memcached	Design a distributed in-memory caching layer to reduce database load. Covers consistent hashing, eviction policies (LRU), and thundering herd problems.	Introduction to Memcached (Slideshare)

15	● Design a Recommendation System like Amazon's	Design a product recommendation engine using collaborative filtering and content-based filtering. Covers feature stores, model serving, and A/B testing.	Recommender Systems – IJCAI Tutorial
16	● Design a URL Shortener like Bit.ly	Design a URL shortening service that encodes long URLs, redirects users, and tracks analytics. Covers Base62 encoding, hash collisions, and read-heavy caching.	System Design for Big Data – TinyURL
17	● Design a Chat App like WhatsApp	Design a real-time 1:1 and group messaging application with delivery receipts, end-to-end encryption, and offline queuing. Covers WebSocket, message storage, and fanout.	WhatsApp Architecture – High Scalability
18	● Design a Photo Sharing System like Instagram	Design a service to upload, store, filter, and serve billions of photos with a social feed. Covers object storage, CDN, feed generation, and media processing pipelines.	Instagram Architecture – High Scalability
19	● Design Facebook News Feed	Design a personalized activity feed showing posts, likes, and comments from friends. Covers fan-out on write vs. read, feed ranking, and real-time updates.	Quora – Best Practices for a News Feed
20	● Design Facebook Timeline	Design a chronological profile timeline that aggregates stories, photos, and life events efficiently at scale. Covers denormalization and aggregation pipelines.	Facebook Engineering – Building Timeline
21	● Design Facebook Chat	Design the real-time messaging backbone behind Facebook Chat supporting billions of users. Covers Erlang-based message routing, presence, and push delivery.	Erlang at Facebook – Erlang Factory
22	● Design a Graph Search like Facebook Graph Search	Design a natural-language search system over a social graph to answer queries like "Friends who work at Google." Covers indexing, entity extraction, and graph traversal.	Facebook Engineering – Building Graph Search Infrastructure
23	● Design a CDN like Cloudflare	Design a globally distributed content delivery network that serves static/dynamic content from edge nodes close to users. Covers Anycast routing, cache invalidation, and DDoS protection.	CMU – Globally Distributed Content Delivery

24	● Design a Trending Topics System like Twitter Trends	Design a real-time system that detects and ranks trending hashtags or topics from a high-velocity stream of events. Covers sliding windows, approximate counting (Count-Min Sketch).	Implementing Real-Time Trending Topics in Storm
25	● Design a Unique ID Generator (like Twitter Snowflake)	Design a distributed ID generation service that produces globally unique, time-sortable 64-bit IDs without a single point of coordination. Covers epoch timestamps, machine IDs, and sequence numbers.	Twitter Engineering – Announcing Snowflake
26	● Design a Top-K Frequency Counter (Sliding Window)	Design a system that returns the top-K most frequent items (e.g., search queries) during any given time window in real time. Covers approximate algorithms and streaming data structures.	Efficient Computation of Top-k Elements in Data Streams (Paper)
27	● Design a Multi-Datcenter Data Serving System	Design a system that reads and writes data across multiple geographically distributed datacenters while maintaining consistency and low latency. Covers replication, conflict resolution, and routing.	How Google Serves Data from Multiple Datacenters – High Scalability
28	● Design an Online Multiplayer Card Game	Design a real-time multiplayer card game backend that handles matchmaking, game state synchronization, and latency tolerance. Covers WebSockets, state machines, and async updates.	Real-Time Multiplayer in HTML5
29	● Design a Garbage Collection System	Design a basic tracing garbage collector that manages memory allocation, object graph traversal, and reclamation. Covers mark-and-sweep, generational GC concepts.	Baby's First Garbage Collector – stuffwithstuff.com
30	● Design an API Rate Limiter	Design a rate limiting service that throttles API requests per user/IP using token bucket or leaky bucket algorithms. Covers distributed counters, Redis sliding windows, and rejection strategies.	Stripe Engineering – Rate Limiting
31	● Design a Stock Exchange like NASDAQ	Design a high-frequency trading exchange with an order book, matching engine, and real-time trade execution. Covers ultra-low latency, in-memory order books, and event sourcing.	Jane Street – Exchange Design (YouTube)

32	● Design Uber / Lyft (Ride-Sharing Platform)	Design the backend for a ride-hailing platform with real-time driver matching, geolocation tracking, ETA calculation, and surge pricing. Covers geohashing and real-time messaging.	How Uber Scales Their Real-Time Market Platform – High Scalability
33	● Design Netflix (Video Streaming Service)	Design a large-scale video streaming platform with encoding pipelines, adaptive bitrate delivery, personalized recommendations, and global CDN. Covers chunked video, microservices, and chaos engineering.	A 360 Degree View of the Netflix Stack – High Scalability
34	● Design a Distributed Message Queue like Apache Kafka	Design a high-throughput, fault-tolerant pub/sub message queue used for event streaming and decoupling services. Covers partitions, consumer groups, offsets, and replication.	Kafka Architecture – Slideshare
35	● Design Google Maps (Mapping & Navigation)	Design a mapping service with routing, real-time traffic, and turn-by-turn navigation for billions of users. Covers graph search (Dijkstra/A*), tile-based map rendering, and location indexing.	Scalable Web Architecture and Distributed Systems – AOSA
36	● Design a Distributed File System like HDFS/GFS	Design a fault-tolerant distributed file system optimized for large sequential reads and batch processing workloads. Covers namenode, datanodes, replication, and chunk servers.	Google File System – Google Research Paper
37	● Design a Real-Time Leaderboard System	Design a global leaderboard that ranks millions of users in real time with low-latency reads and frequent score updates (e.g., gaming or coding competitions). Covers sorted sets (Redis ZSET).	Introduction to Redis – Slideshare
38	● Design an E-Commerce Platform like Amazon	Design a large-scale online marketplace supporting product catalog, inventory management, shopping cart, checkout, and order tracking at massive scale. Covers service decomposition and eventual consistency.	Amazon Architecture – High Scalability

39	● Design a Hotel / Flight Booking System	Design a booking platform with availability checks, seat/room reservations, payment flows, and overbooking prevention. Covers distributed transactions, idempotency, and two-phase commit.	Booking.com Development Blog
40	● Design a Typeahead / Autocomplete System	Design a search suggestion service that returns real-time prefix completions as users type. Covers trie data structures, distributed prefix trees, and caching hot prefixes.	Implementing Search Engines – Arden Dertat
41	● Design a Video Conferencing System like Zoom	Design a low-latency, real-time video and audio conferencing system supporting thousands of concurrent meetings. Covers WebRTC, SFU vs. MCU topology, TURN/STUN servers, and packet loss recovery.	How Zoom Works – High Scalability
42	● Design a Distributed Lock Service like ZooKeeper	Design a coordination service for distributed lock management, leader election, and configuration storage. Covers ZAB consensus protocol, ephemeral nodes, and watch callbacks.	Introduction to Apache ZooKeeper – Slideshare
43	● Design a Payment Processing System	Design a reliable payment gateway that handles authorization, settlement, fraud checks, and idempotent retries. Covers ACID transactions, idempotency keys, and PCI-DSS compliance patterns.	Stripe Engineering Blog
44	● Design a Proximity / Location-Based Search (like Yelp)	Design a service that returns nearby businesses or points of interest given a user's coordinates. Covers geohashing, quadrees, and spatial indexing.	System Design Primer – Database Sharding
45	● Design a Push Notification System	Design a notification dispatch system that delivers millions of targeted push notifications (mobile, email, SMS) reliably and at low latency. Covers APNs/FCM, fan-out queues, and deduplication.	Delivering Billions of Messages Instantly – Facebook Engineering
46	● Design a Live Video Streaming Service like Twitch	Design a live streaming platform that ingests real-time video, transcodes it, and delivers it to millions of concurrent viewers. Covers RTMP ingest, HLS/DASH delivery, and CDN edge caching.	Justin.TV Live Video Broadcasting Architecture – High Scalability

47	● Design an Email Service like Gmail	Design a scalable email system with sending, receiving, indexing, threading, spam filtering, and storage for billions of messages. Covers SMTP relay, IMAP, full-text search, and tiered storage.	Scaling Gmail – Google Talks (YouTube)
48	● Design a Distributed Database like Google Spanner	Design a globally distributed, strongly consistent SQL database. Covers TrueTime API, Paxos consensus per shard, external consistency, and 2PC between shard groups.	Spanner: Google's Globally-Distributed Database (Paper)
49	● Design MapReduce	Design the MapReduce distributed computation framework for processing large datasets in parallel across commodity hardware. Covers map/shuffle/reduce phases, fault tolerance, and stragglers.	MapReduce: Simplified Data Processing on Large Clusters (Google Paper)
50	● Design a Log Aggregation / Monitoring System like Splunk	Design a platform to ingest, parse, index, and query high-volume log data from thousands of services in near real-time. Covers log shippers, indexing pipelines, and time-series storage.	ESPN Architecture at Scale – High Scalability
51	● Design a Blogging Platform like Medium	Design a publishing platform supporting rich-text articles, comments, claps, tags, and personalized reading feeds for millions of writers and readers. Covers CMS, full-text search, and CDN.	Medium Engineering Blog
52	● Design a Social Network like LinkedIn	Design a professional social network with profile pages, connections, messaging, job listings, and a feed. Covers graph storage, InMail, and search over structured profiles.	LinkedIn Engineering Blog
53	● Design a Code Repository Hosting Service like GitHub	Design a service for hosting Git repositories with access control, pull requests, code review, CI/CD integration, and issue tracking at scale. Covers Git object storage and webhook delivery.	GitHub Engineering Blog
54	● Design a Document Search & Indexing System like Elasticsearch	Design a distributed, schema-free search and analytics engine for full-text and structured queries. Covers inverted index, sharding, relevance scoring (BM25), and near real-time indexing.	Elasticsearch: From the Bottom Up (Elastic Blog)

55	● Design a Consistent Hashing Ring	Design a data partitioning scheme using a consistent hash ring that minimizes key remapping when nodes join or leave. Covers virtual nodes, replication factor, and token-based ranges.	Consistent Hashing – Tom White
56	● Design a Distributed Consensus System (Paxos / Raft)	Design or explain a distributed consensus algorithm ensuring all nodes in a cluster agree on a single value even under network partitions and node failures. Covers leader election and log replication.	Paxos Made Simple – Microsoft Research
57	● Design a Food Delivery App like DoorDash / Uber Eats	Design a food delivery marketplace connecting customers, restaurants, and couriers with real-time order tracking and ETA prediction. Covers location services, dynamic dispatch, and order state machines.	DoorDash Engineering Blog
58	● Design a Music Streaming Service like Spotify	Design a music streaming service with catalog management, personalized playlists, audio streaming, and offline downloads. Covers content licensing, adaptive streaming, and recommendation models.	Spotify Labs Engineering
59	● Design a Ticket Booking System like Ticketmaster	Design a high-demand event ticketing system that handles flash sales, prevents overselling, and queues users fairly under massive load spikes. Covers seat locking, virtual queues, and idempotent purchases.	Handling Flash Sales at Scale – High Scalability
60	● Design Cloud Object Storage like AWS S3	Design a scalable, durable, highly available object storage service. Covers consistent hashing for object placement, erasure coding, multipart uploads, and eventual vs. strong consistency.	Dynamo: Amazon's Highly Available Key-Value Store (Paper)
61	● Design a Feature Flag / A/B Testing Platform	Design a platform that enables controlled rollout of features and statistical A/B experiments across services without redeployments. Covers flag evaluation, assignment hashing, and metric collection.	Airbnb Engineering – Experimentation at Airbnb
62	● Design a Fraud Detection System	Design a real-time system that scores and blocks fraudulent transactions using rules engines and ML models. Covers feature engineering, low-latency scoring, and feedback loops.	Uber Engineering – Mastermind: Uber's Fraud Detection System

63	● Design a Job Scheduling System (like Airflow / Cron)	Design a distributed task scheduler that runs recurring or event-driven jobs reliably, with retries and observability. Covers DAG scheduling, worker pools, and distributed locking.	Airflow Architecture – Apache Airflow Docs
64	● Design a Real-Time Analytics Pipeline	Design a streaming analytics system that ingests events, aggregates metrics, and surfaces dashboards with sub-second latency. Covers Kafka → Flink/Spark Streaming → time-series DB.	Apache Flink Architecture
65	● Design an Ad Click-Through Rate (CTR) Prediction System	Design a machine learning pipeline that predicts the probability a user clicks an ad, and serves predictions at millisecond latency under billions of daily requests. Covers feature stores and model serving.	LinkedIn – Scaling Distributed Machine Learning with the Parameter Server
66	● Design a DNS System	Design a globally distributed hierarchical Domain Name System that resolves domain names to IP addresses with caching, TTL, and authoritative/recursive resolvers.	DNS Architecture – Microsoft TechNet
67	● Design a Health Monitoring & Alerting System	Design an observability platform that collects metrics from millions of services, evaluates alert rules in real time, and notifies on-call engineers. Covers time-series storage, anomaly detection, and silence/routing.	Netflix Engineering – Introducing Atlas
68	● Design a Bigtable / Wide-Column Store	Design a distributed, sparse, sorted wide-column store optimized for large-scale structured data. Covers SSTables, LSM trees, compaction, and tablet servers.	Bigtable: A Distributed Storage System (Google Paper)
69	● Design a Ride-Matching and ETA Service	Design the matching engine that pairs riders with the nearest available drivers and computes accurate ETAs. Covers geohashing, supply/demand modeling, and real-time map routing.	Uber Engineering Blog
70	● Design a Content Moderation System	Design an automated system that detects and removes harmful content (spam, hate speech, CSAM) from a social platform at scale. Covers ML classifiers, human review queues, and appeals workflows.	Facebook Engineering – Scalable Spam Fighting

71	● Design a CI/CD Pipeline (Code Deployment System)	Design a continuous integration and delivery system that builds, tests, and deploys code changes automatically and reliably. Covers build isolation, artifact storage, rollout strategies, and rollback.	LinkedIn Engineering – Continuous Delivery
72	● Design an Online Judge like LeetCode	Design a system that accepts user-submitted code, compiles and executes it in a sandbox, and evaluates it against test cases. Covers sandboxing, resource limits, and anti-cheating.	System Design Primer – Application Layer
73	● Design a Collaborative Whiteboard / Figma	Design a real-time collaborative design tool supporting shared canvases, vector graphics, and simultaneous multi-user edits. Covers CRDTs, WebSocket broadcasting, and conflict-free merges.	Figma Engineering – How Figma's Multiplayer Technology Works
74	● Design a Hotel Review Platform like TripAdvisor	Design a platform handling millions of reviews, photos, and rankings for hotels and attractions with search and personalization. Covers read-heavy caching, full-text indexing, and UGC pipelines.	TripAdvisor Architecture – High Scalability
75	● Design a Distributed Tracing System like Dapper / Zipkin	Design a system that traces requests across hundreds of microservices, correlates spans, and helps engineers debug latency and errors. Covers trace context propagation, sampling strategies, and storage.	Dapper: A Large-Scale Distributed Systems Tracing Infrastructure (Google Paper)

Quick Reference by Category

Category	Question Numbers
Storage & Databases	6, 13, 14, 36, 48, 49, 60, 68
Social & Feeds	2, 5, 18, 19, 20, 21, 22, 52, 53
URL / ID Generation	1, 16, 25
Search & Indexing	10, 11, 40, 54, 74
Messaging & Streaming	17, 21, 34, 45, 46, 47
Caching	6, 14, 37, 50
Distributed Systems	27, 42, 49, 55, 56, 66, 75

Real-Time / Events	24, 26, 28, 64
E-Commerce & Booking	7, 38, 39, 57, 59
Infra & DevOps	9, 23, 30, 60, 63, 67, 71
Ride-Sharing / Maps	32, 35, 41, 69
ML & Recommendations	15, 61, 62, 65
Media & Entertainment	18, 33, 46, 58, 73
Finance & Payments	31, 43
Crawlers & Indexing	3, 10, 11

Key Resources

Resource	Link
System Design Primer (GitHub)	https://github.com/donnemartin/system-design-primer
checkcheckzz / System Design Interview	https://github.com/checkcheckzz/system-design-interview
High Scalability Blog	http://highscalability.com/
Grokking the System Design Interview	https://www.educative.io/courses/grokking-the-system-design-interview
System Design Interview (Book) by Alex Xu	https://www.amazon.com/System-Design-Interview-insiders-Second/dp/B08CMF2CQF
Designing Data-Intensive Applications	https://www.amazon.com/dp/1449373321
MIT 6.824 Distributed Systems (Free Course)	https://www.youtube.com/playlist?list=PLrw6a1wE39_tb2fErI4-WkMbsvGQk9_UB
Google Research Papers Index	https://research.google/pubs/